

Waves • Student Workbook

Name: _____ Class: _____

1 • Wave basics

1. Explain the statement: “waves transfer energy without transferring matter.”

2. Compare transverse and longitudinal waves. Include a labelled sketch.

2 • Equations

$$T = 1/f \quad v = f\lambda$$

3. A wave has $f = 8 \text{ Hz}$. Find T .

4. A wave has $f = 12 \text{ Hz}$ and $\lambda = 0.40 \text{ m}$. Find v .

3 • Ripple tank

5. Draw and explain reflection, refraction and diffraction of plane water waves.

6. Explain why a stroboscope is useful.

4 • Sound

7. Explain the bell-jar experiment and what it shows.

8. An echo returns after 0.60 s. Speed of sound = 340 m/s. Find the distance to the reflecting wall.

9. State how frequency and amplitude affect what we hear.

5 • EM waves and light

10. Write the seven EM regions in increasing wavelength.

11. Draw a reflection ray diagram and label normal, i and r.

12. State the two conditions for total internal reflection.

6 • Lenses

13. Complete an image-properties table for object beyond $2F$, at $2F$, between F and $2F$, and inside F .

14. An image is 9 cm high and the object is 3 cm high. Find magnification.

Waves • Student Activities

Fun practicals, PhET investigations, video thinking routines and team challenges linked directly to the presentation.

Activity Lab

These are designed for mixed-ability 13–14-year-olds. The rule is **predict** → **test** → **evidence** → **explain**. Do not skip the prediction: being wrong and then changing your mind is part of the physics.

Teacher: full apparatus, setup, safety, differentiation and questioning notes are in the Teacher Tools guide.

Activity 1

Foundations

8 min

Human Wave: Energy Moves, People Stay

What you need

Open space; 8–12 students; one coloured wristband or sticky note.

Student mission

1. Student 1 raises and lowers both hands once.
2. Each student copies only when the neighbour's hands rise.
3. Repeat faster, then reverse direction.
4. Track the marked student: they move locally but do not travel with the pulse.

Think like a physicist

- Predict: will the marked student reach the other end?
- What moved across the line: people, energy, or both?
- What would have to change for matter to be transferred?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Slinky Particle Detective

What you need

Long slinky; masking tape; floor markers.

Student mission

1. Make a transverse pulse by moving one end sideways once.
2. Watch only the taped coil.
3. Make a longitudinal pulse by pushing/pulling along the slinky.
4. Students sketch particle motion and energy direction for both.

Think like a physicist

- Predict the taped coil's motion before each pulse.
- What evidence makes one wave transverse and the other longitudinal?
- If the pulse travels right, must the coil travel right?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Frequency–Period Beat Challenge

What you need

Stopwatch; metronome app optional; mini-whiteboards.

Student mission

1. Teacher gives a steady beat for 10 s.
2. Pairs count cycles and estimate frequency.
3. Use $T = 1/f$ to predict period.
4. Check by timing 5 or 10 cycles and dividing.

Think like a physicist

- Which is more reliable: timing one cycle or ten? Why?
- If frequency doubles, what happens to period?
- Can two waves have the same period but different frequency?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

PhET Wave Detective Lab

What you need

Projector or student devices;
internet; PhET Wave on a String;
worksheet.

Student mission

1. Predict what increasing frequency will change.
2. Test and measure wavelength with the ruler.
3. Reset; change amplitude only.
4. Reset; change tension and compare wave speed.
5. Write one cause–effect rule supported by evidence.

Think like a physicist

- What stayed constant in each test?
- Why is “the wave looks faster” weaker evidence than a measurement?
- Can amplitude change without changing frequency?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Stroboscope Freeze Challenge

What you need

Ripple tank; wave generator; stroboscope/strobe control; ruler.

Student mission

1. Observe without the strobe.
2. Predict what the strobe changes physically.
3. Adjust until wavefronts appear frozen.
4. Measure wavelength more easily.
5. Explain why the water itself has not stopped.

Think like a physicist

- Does the stroboscope change the wave or only what we see?
- Why does a stationary-looking pattern help measurement?
- What would happen if the strobe frequency were slightly wrong?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Reflection Detective

What you need

Ripple tank; straight dipper; straight barrier; lamp; paper; ruler; protractor.

Student mission

1. Draw a normal at the reflection point.
2. Measure angle of incidence from the normal.
3. Predict reflection angle.
4. Trace reflected wavefronts and measure.
5. Repeat for three incidence angles.

Think like a physicist

- Why measure from the normal rather than the barrier?
- What evidence supports $i = r$?
- Which variable should stay unchanged across trials?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Deep-to-Shallow Refraction Mystery

What you need

Ripple tank; straight dipper; submerged glass/acrylic plate; lamp; ruler; protractor.

Student mission

1. Predict speed, wavelength and direction in shallow water.
2. Observe wavefront spacing before/after boundary.
3. Trace rays perpendicular to wavefronts.
4. Repeat at normal incidence.
5. Explain why frequency stays unchanged.

Think like a physicist

- Which observation tells you speed changed?
- Why can wavelength change while frequency stays fixed?
- Why is there no bending at normal incidence?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Diffraction Gap Tournament

What you need

Ripple tank; two barriers; ruler; straight dipper; optional PhET Wave Interference.

Student mission

1. Keep wavelength fixed; test a wide gap and narrow gap.
2. Rank spreading from least to most.
3. Keep gap fixed; change wavelength.
4. Draw wavefronts for two contrasting cases.
5. Team must defend its rule with evidence.

Think like a physicist

- What ratio matters: gap width compared with what?
- How would you make diffraction stronger without changing the gap?
- Where else could waves bend around obstacles?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Cup Telephone Investigation

What you need

Paper cups; string; paper clips; ruler/tape; optional different strings.

Student mission

1. Test with string taut, then slack.
2. Test touching vs not touching the string.
3. If time permits compare string length/material.
4. Record which setup transmits sound best.
5. Explain using vibrations and a material medium.

Think like a physicist

- Where does the vibration start?
- What carries energy between cups?
- Why does slack string reduce transmission?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Tuning Fork Splash Proof

What you need

Tuning fork; rubber striker; shallow tray/beaker of water; paper towel.

Student mission

1. Predict what the water will do.
2. Strike the fork and listen.
3. Touch prongs lightly to water.
4. Link splashes to invisible vibration.
5. Repeat with a different frequency fork if available.

Think like a physicist

- What evidence shows the fork is vibrating?
- Does a louder sound necessarily mean a higher frequency?
- What changes if the fork is struck harder?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Scream in Space: Claim–Evidence–Reasoning

What you need

Included MythBusters Jr. clip; mini-whiteboards or paper.

Student mission

1. Write CLAIM: can sound cross a vacuum?
2. Write a reason using particles/medium.
3. Watch the relevant clip segment.
4. Add EVIDENCE from the observation.
5. Rewrite the explanation as Claim–Evidence–Reasoning.

Think like a physicist

- What result would falsify your original claim?
- Is seeing vibration enough to prove sound travelled?
- Why is “space is quiet” not itself a mechanism?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Pitch vs Loudness Mystery

What you need

CRO Flash/PhET Waves Intro Sound; speaker at low volume; mini-whiteboards.

Student mission

1. Students vote: pitch changed, loudness changed, both, or neither.
2. Reveal waveform.
3. Identify frequency and amplitude separately.
4. Students create a “trick pair” for another team.

Think like a physicist

- Can a quiet sound have high frequency?
- Can a loud sound have low frequency?
- Which graph feature is evidence for each judgement?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Bat Sonar Distance Challenge

What you need

Included echolocation video; calculators; ruler; scenario cards.

Student mission

1. Identify that measured time is out-and-back.
2. Calculate total path using $s=vt$.
3. Halve to find target distance.
4. Compare team scenarios.
5. Explain why shorter echo time means a nearer object.

Think like a physicist

- Why must distance be halved?
- What assumption are we making about sound speed?
- What could make the calculated distance inaccurate?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

EM Spectrum Human Line-Up

What you need

Seven printed spectrum cards; use/source cards; open space.

Student mission

1. First attempt from memory.
2. Class may give only one-word clues.
3. Add use/source cards and match them.
4. Reverse order for increasing frequency.
5. Explain why wavelength order reverses frequency order.

Think like a physicist

- Which end has highest frequency?
- Which regions are ionising at the high-frequency end?
- Why does changing order from wavelength to frequency reverse the line?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Mirror Maze

What you need

Ray boxes; plane mirrors; white paper; rulers; protractors; target card.

Student mission

1. Predict mirror position before switching on ray box.
2. Draw normals at reflection points.
3. Place mirror and test.
4. Measure i and r .
5. Modify maze with a second reflection.

Think like a physicist

- Can you place the mirror using geometry before testing?
- What evidence shows the law of reflection holds at both mirrors?
- How is this related to a periscope?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Coin Reappears: Refraction Prediction

What you need

Opaque cup/bowl; coin; water jug; towel.

Student mission

1. Predict whether the coin will remain hidden.
2. Add water slowly.
3. Observe the coin reappear.
4. Draw rays bending at the water–air boundary.
5. Explain apparent depth.

Think like a physicist

- Did the coin physically move?
- Where did the light ray change direction?
- Why does the coin appear shallower?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Bending Light Evidence Hunt

What you need

Projector/student devices; internet; PhET Bending Light; worksheet.

Student mission

1. Predict which medium bends light more.
2. Measure i and r for water.
3. Reset and repeat for glass with same i .
4. Use evidence to rank refractive index.
5. Test normal incidence as a control case.

Think like a physicist

- Why must incidence angle be the same for a fair comparison?
- What does a smaller refraction angle suggest?
- Can speed change without direction changing? When?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Total Internal Reflection Hunt

What you need

Semicircular acrylic block; ray box; paper; protractor.

Student mission

1. Start below the critical angle.
2. Increase incidence angle gradually.
3. Find the angle where refracted ray is along the boundary.
4. Increase further and observe TIR.
5. Write the two conditions for TIR.

Think like a physicist

- Why must light travel from denser to less dense medium?
- What changes exactly at the critical angle?
- How does this make optical fibres possible?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Lens Image Scavenger Hunt

What you need

Converging lens; lens holder; illuminated object/window; white screen; metre rule; optional PhET Geometric Optics.

Student mission

1. Find a sharp real image on the screen.
2. Move object to make image larger.
3. Move object to make image smaller.
4. Try object inside F and explain why screen image disappears.
5. Record object position and image properties.

Think like a physicist

- What evidence tells you an image is real?
- Why can a virtual image not be caught on the screen?
- Where should the object be for a magnified real image?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Waves Escape Room: Six Locks

What you need

Printed lock cards from Student Activities page; calculators; mini-whiteboards; optional envelopes.

Student mission

1. Give each team Lock 1 only.
2. Require reasoning, not just a number.
3. Wrong answer → one hint after explanation attempt.
4. Finish with a team explanation of the hardest lock.

Think like a physicist

- Which clue made you change strategy?
- How did you decide which equation or rule applied?
- What mistake would produce the most tempting wrong answer?

Record

Write your prediction first, then your observation/evidence, then your explanation. If your prediction was wrong, explain what changed your mind.

Waves • Assessment Book

Six part-assessments plus a final mixed quiz, aligned to the presentation sequence and MATSEC 2027 Waves content.

How to use this book

Complete each part after that block of the topic. Each section mixes quick recall with explanation, calculations and practical thinking. Use **Print** for paper copies, or type directly into the boxes. The final quiz is a 20-question mixed check.

Suggested target: 70%+ on the multiple-choice checks before moving on; then correct written answers using teacher feedback.

Part 1 • Wave foundations

Sections 1–5: energy transfer, transverse/longitudinal waves, amplitude, wavelength, frequency, period, wave speed and graphs.

1. A wave transfers...

- ☐ A. matter only
- ☐ B. energy without overall transfer of matter
- ☐ C. particles from source to receiver
- ☐ D. mass and energy together

2. In a transverse wave, particle vibration is...

- ☐ A. parallel to energy transfer
- ☐ B. perpendicular to energy transfer
- ☐ C. always vertical
- ☐ D. in the same direction as wave speed

3. A wave has frequency 5 Hz. Its period is...

- ☐ A. 0.20 s
- ☐

☐ B. 5 s

☐ C. 2.5 s

☐ D. 25 s

4. A wave has $f = 8 \text{ Hz}$ and $\lambda = 0.50 \text{ m}$. Its speed is...

☐ A. 4 m/s

☐ B. 16 m/s

☐ C. 8.5 m/s

☐ D. 0.0625 m/s

5. Explain the difference between a displacement–distance graph and a displacement–time graph.

6. A wave travels at 12 m/s and has frequency 3 Hz. Calculate its wavelength. Show equation, substitution and unit.

7. Draw or describe one wavelength on a longitudinal wave.



Part 2 • Ripple tank & wave behaviour

Sections 6–9: wavefronts, stroboscope, reflection, refraction and diffraction.

1. A ray on a wave diagram is drawn...

- ☐ A. parallel to the wavefront
- ☐ B. perpendicular to the wavefront
- ☐ C. only for light
- ☐ D. along a crest

2. When water waves enter shallower water, their frequency...

- ☐ A. increases
- ☐ B. decreases
- ☐ C. stays the same
- ☐ D. becomes zero

3. Diffraction is strongest when the gap is...

- ☐ A. much wider than λ
- ☐ B. comparable to or smaller than λ

☐ C. always exactly 1 m

☐ D. at normal incidence

4. For reflection from a straight barrier, the angle of reflection is...

☐ A. twice i

☐ B. equal to i

☐ C. $90^\circ - i$

☐ D. always zero

5. Why is a stroboscope useful in a ripple-tank experiment?

6. Explain why wavelength decreases when a water wave enters shallower water while frequency stays constant.

7. Design a short investigation to test how gap width affects diffraction. Include one control variable and what you would observe.

Part 3 • Sound & ultrasound

Sections 10–12: sound as a mechanical longitudinal wave, speed, echoes, pitch, loudness and ultrasound.

1. Sound in air is...

- ☐ A. transverse electromagnetic
- ☐ B. longitudinal mechanical
- ☐ C. transverse mechanical
- ☐ D. longitudinal electromagnetic

2. Increasing sound frequency mainly increases...

- ☐ A. loudness
- ☐ B. pitch
- ☐ C. amplitude
- ☐ D. speed in the same medium

3. Ultrasound has frequency...

- ☐ A. below 20 Hz
- ☐

B. above 20 000 Hz

☐

C. exactly 340 Hz

☐

D. between 20 and 200 Hz

4. An echo returns after 0.40 s. Take speed of sound as 340 m/s. Find the distance to the wall.

5. Explain the bell-jar experiment evidence that sound needs a medium.

6. Distinguish pitch from loudness using wave quantities.

7. Explain how sonar can measure depth.

Part 4 • Electromagnetic spectrum

Section 13: common properties, order, uses, sources/detectors and visible light.

1. Which EM wave has the longest wavelength?

- ☐ A. gamma
- ☐ B. X-ray
- ☐ C. visible
- ☐ D. radio

2. All electromagnetic waves can...

- ☐ A. travel only in solids
- ☐ B. travel through a vacuum
- ☐ C. be heard by humans
- ☐ D. move slower than sound

3. As EM frequency increases, wavelength in vacuum...

- ☐ A. increases
- ☐ B. decreases

☐

C. stays constant

☐

D. becomes unrelated

4. Write the seven EM regions in order of increasing wavelength.

5. Give one use of infrared and explain the property that makes it useful.

6. An EM wave has frequency 1.0×10^8 Hz. Use $c = 3.0 \times 10^8$ m/s to calculate wavelength.

7. Put visible colours in order from longest to shortest wavelength.

Part 5 • Reflection, refraction & TIR

Sections 14–16: rays, plane mirrors, refraction, refractive index, apparent depth, critical angle and fibres.

1. A ray hits a mirror at 35° to the normal. Reflection angle is...

- ☐ A. 35°
- ☐ B. 55°
- ☐ C. 70°
- ☐ D. 145°

2. Light entering glass from air at an angle normally bends...

- ☐ A. away from the normal
- ☐ B. towards the normal
- ☐ C. along the boundary
- ☐ D. back to the source

3. At normal incidence, a ray entering glass...

- ☐ A. changes speed but not direction
- ☐

B. changes direction but not speed

☐

C. undergoes TIR

☐

D. must reflect completely

4. State two properties of an image in a plane mirror.

5. For a material, $\sin i = 0.75$ and $\sin r = 0.50$. Calculate refractive index $n = \sin i / \sin r$.

6. State both conditions needed for total internal reflection.

7. Explain why optical fibres can guide light around bends.

Part 6 • Dispersion & lenses

Section 17: dispersion, converging lenses, principal rays, image properties, magnification and focal length.

1. A converging lens is...

- ☐ A. thinner in the middle
 - ☐ B. thicker in the middle
 - ☐ C. always a plane mirror
 - ☐ D. unable to form real images
-

2. For an object beyond $2F$, a converging lens forms an image that is...

- ☐ A. real, inverted, smaller
 - ☐ B. virtual, upright, larger
 - ☐ C. real, upright, same size
 - ☐ D. virtual, inverted, smaller
-

3. For an object inside F , the image is...

- ☐ A. real and inverted
- ☐

B. virtual, upright and magnified

☐

C. real and same size

☐

D. at $2F$

4. Name the three principal rays commonly used to construct a converging-lens ray diagram.

5. An object is 4 cm high and its image is 10 cm high. Calculate magnification.

6. Why does a prism disperse white light?

7. Describe how to measure the focal length of a converging lens using a distant object.

Final assessment

20 questions

Final Waves Quiz

Do this without notes first. Then use your errors to choose what to revise.

1. Which statement best describes a wave?

- ☐ A. A transfer of energy without overall transfer of matter
- ☐ B. A transfer of matter only
- ☐ C. A permanent movement of particles with the wave
- ☐ D. A force that only exists in water

2. A particle vibrates parallel to the direction of energy transfer. The wave is...

- ☐ A. transverse
- ☐ B. longitudinal
- ☐ C. electromagnetic only
- ☐ D. stationary

3. A wave has $f=4$ Hz. Its period is...

- ☐ A. 0.25 s
- ☐ B. 4 s

☐
C. 2 s

☐
D. 8 s

4. A wave has $v=6$ m/s and $f=3$ Hz. Its wavelength is...

☐
A. 18 m

☐
B. 9 m

☐
C. 2 m

☐
D. 0.5 m

5. Water waves entering shallow water normally...

☐
A. speed up and λ increases

☐
B. slow down and λ decreases

☐
C. keep speed but change f

☐
D. stop at the boundary

6. Diffraction becomes stronger when...

☐
A. gap becomes large compared with λ

☐
B. gap becomes comparable with λ

☐

☐ C. frequency is always zero

☐ D. reflection angle increases

7. Sound cannot travel through...

☐ A. air

☐ B. water

☐ C. steel

☐ D. a vacuum

8. A larger sound-wave amplitude is mainly heard as...

☐ A. higher pitch

☐ B. greater loudness

☐ C. higher speed

☐ D. shorter period

9. An echo time is 0.60 s and sound speed is 340 m/s. Wall distance is...

☐ A. 204 m

☐ B. 102 m

☐ C. 340 m

- ☐
- D. 56.7 m

10. Ultrasound is sound above approximately...

- ☐ A. 20 Hz
- ☐ B. 200 Hz
- ☐ C. 2 000 Hz
- ☐ D. 20 000 Hz

11. Which order goes from longest to shortest wavelength?

- ☐ A. radio → microwave → infrared → visible → ultraviolet → X-ray → gamma
- ☐ B. gamma → X-ray → ultraviolet → visible → infrared → microwave → radio
- ☐ C. radio → infrared → microwave → visible → gamma → X-ray → ultraviolet
- ☐ D. visible → radio → gamma → infrared → X-ray → microwave → ultraviolet

12. All EM waves in vacuum have the same...

- ☐ A. frequency
- ☐ B. wavelength
- ☐ C. speed
- ☐

D. energy

13. A mirror is hit at 28° to the normal. Reflection angle is...

☐ A. 28°

☐ B. 62°

☐ C. 56°

☐ D. 90°

14. A ray entering glass from air obliquely bends...

☐ A. towards the normal

☐ B. away from the normal

☐ C. only if it is red

☐ D. without changing speed

15. At normal incidence into glass, light...

☐ A. changes direction only

☐ B. changes speed but not direction

☐ C. undergoes TIR

☐ D. must stop

16. TIR requires light to travel...

- ☐ A. less dense to denser, any angle
- ☐ B. denser to less dense, $i >$ critical angle
- ☐ C. denser to less dense, $i <$ critical angle
- ☐ D. through a mirror

17. Optical fibres mainly rely on...

- ☐ A. diffraction
- ☐ B. dispersion
- ☐ C. total internal reflection
- ☐ D. sound reflection

18. White light separates in a prism because colours...

- ☐ A. all refract by exactly the same amount
- ☐ B. have different wavelengths and refract by different amounts
- ☐ C. have different speeds in vacuum
- ☐ D. are sound waves

19. Object beyond 2F in a converging lens gives...

- ☐ A. real inverted smaller image
- ☐ B. virtual upright larger image
- ☐ C. real upright image
- ☐ D. no image

20. Image height 12 cm, object height 4 cm. Magnification is...

- ☐ A. 0.33
- ☐ B. 3
- ☐ C. 8
- ☐ D. 48

Waves - New Fun Activities 21-25

These activities are now linked directly from presentation slides 19, 39, 40, 46 and 78.

Activity 21: Wave Label Relay (Slide 19)

What you need

Mini-whiteboards/A4 paper; coloured pens; slide 19 video.

Student mission

1. Watch a short segment, then pause.
2. Draw one wave from memory.
3. Label crest, trough, equilibrium, amplitude and wavelength.
4. Swap boards and check another team.

Think like a physicist

- Why is amplitude measured from equilibrium?
- Which matching points can define wavelength?
- Which quantity needs time to measure: frequency or wavelength?

Record: prediction -> evidence -> explanation. Correct one idea if the evidence changes your mind.

Activity 22: Ripple Tank Pattern Hunt (Slide 39)

What you need

Slide 39 virtual ripple tank; worksheet or mini-whiteboard.

Student mission

1. Predict straight-dipper pattern.
2. Switch to point source and sketch it.
3. Increase frequency at fixed depth.
4. Pause and compare wavefront spacing.

Think like a physicist

- What does each blue line represent?
- Why do closer fronts mean shorter wavelength?
- What evidence links source shape to wavefront shape?

Record: prediction -> evidence -> explanation. Correct one idea if the evidence changes your mind.

Activity 23: Human Wavefront and Ray (Slide 40)

What you need

Masking tape/floor line; 5-8 students; arrow card.

Student mission

1. Students form one straight wavefront.
2. Another student holds the energy-direction arrow.
3. Rotate the wavefront and place the ray at 90 degrees.
4. Check against slide 40.

Think like a physicist

- Is a ray a physical line in water?
- What information does a ray represent?
- Why is it perpendicular to a plane wavefront?

Record: prediction -> evidence -> explanation. Correct one idea if the evidence changes your mind.

Activity 24: Beat the Reflection Simulator (Slide 46)

What you need

Slide 46 reflection animation; protractor; mini-whiteboards.

Student mission

1. Teacher chooses an incidence angle.
2. Teams predict r before running.
3. Pause at the barrier and identify normal, i and r .
4. Repeat for three angles.

Think like a physicist

- Why measure i and r from the normal?
- How are wavefronts oriented relative to rays?
- What evidence supports $i = r$?

Record: prediction -> evidence -> explanation. Correct one idea if the evidence changes your mind.

Activity 25: Resonance Detective: Why Did the Glass Break? (Slide 78)

What you need

Slide 78 slow-motion glass video; mini-whiteboards.

Student mission

1. Predict what must match for strong vibration.
2. Watch for visible flexing.
3. Identify evidence of increasing amplitude.
4. Write Claim-Evidence-Reasoning.

Think like a physicist

- Would any frequency work equally well?
- Why does repeated energy transfer matter?
- How is resonance different from simply making a sound louder?

Record: prediction -> evidence -> explanation. Correct one idea if the evidence changes your mind.